

SANA AMBREEN

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EXPERIENCE SUMMARY

Senior GenAI Engineer with 7+ years of experience building and shipping end-to-end generative AI and machine learning solutions for SaaS and enterprise environments. Specialized in LLM application development, multi-agent orchestration, Graph RAG, LangGraph pipelines, and full-stack AI serving across GCP, AWS, and Azure. Led the design and production deployment of a personalized AI nudge engine that increased user engagement by 20-25%, built a real-time trading analytics platform that reduced decision latency by approximately 40% while improving signal precision by 30%, and architected RAG-powered explainability systems with evaluation, guardrails, and cost optimization. Proven ability to translate complex data into actionable insights and communicate findings clearly to both technical and non-technical stakeholders, driving customer success through scalable, measurable AI outcomes.

TECHNICAL SKILLS

Languages: Python, SQL, Bash/Shell, TypeScript, JavaScript, Java, Kotlin

LLM and GenAI: LLMs (Gemini, GPT-4, Llama, BERT), Prompt Engineering (few-shot, zero-shot, chain-of-thought, output parsing), Function/Tool Use, RAG Architectures, Graph RAG, Agentic AI, Multi-Agent Systems, Safety/Guardrails, Evaluation Frameworks, Cost & Latency Optimization, Conversational AI (Dialogflow CX)

Agent and Orchestration Frameworks: LangChain, LangGraph, PydanticAI, Google ADK, CrewAI, Model Context Protocol (MCP), Multi-Agent Orchestration

Vector and Retrieval: FAISS, Pinecone, ChromaDB, pgvector, Vertex AI Matching Engine, text-embedding-gecko, Semantic Similarity Search, Knowledge Graphs

ML and Data: PyTorch, TensorFlow/Keras, Hugging Face, PEFT/LoRA, XGBoost, scikit-learn, Pandas, NumPy, Spark, Hadoop

Cloud and Infra: GCP (Vertex AI, BigQuery, Cloud Run, Firebase), AWS (SageMaker, Lambda, EC2, ECS, SQS), Azure (Synapse, ML Studio, Cognitive Search), Docker, Kubernetes

Engineering and MLOps: REST API Design, FastAPI, Flask, CI/CD (GitHub Actions, Jenkins), MLOps, Model Monitoring, Drift Detection, Observability & Logging, Telemetry, Incident Response

Analytics: Tableau, Power BI, A/B Testing, Statistical Modeling, Time-Series Forecasting, NLP (BERT, BERTweet, FinBERT, SBERT, Text Classification, Sentiment Analysis), Actionable Insights, Generative AI, Customer Success

CAREER SUMMARY

UpLifty AI

GenAI Engineer | Remote (USA) | Oct 2025 - Present

AI-powered wellness and personal growth platform. Owned end-to-end design, deployment, and evaluation of LLM applications, AI agents, and CI/CD infrastructure on GCP.

- Designed and deployed a personalized AI nudge engine (PAI) using event-driven architecture that ingests multi-source behavioral data, evaluates rule-based triggers, and generates grounded recommendations via LLM inference with prompt engineering guardrails and safety filters, improving user engagement by 20-25%.
- Instrumented telemetry and evaluation pipelines to monitor nudge quality, latency, and cost; tracked key performance indicators to maintain system stability and model quality over time.
- Fine-tuned Llama 3.2 (1B) and BERTweet using qLoRA for wellness classification, implementing few-shot prompt templates, chain-of-thought output parsing, and hallucination guardrails to ensure safe, high-quality outputs deployed on Google Cloud Run with sub-300ms inference latency.
- Architected multi-agent orchestration pipelines using LangGraph and LangChain with tool use and function calling, enabling modular agent workflows for personalized recommendation generation, retrieval-augmented grounding, and context-aware response synthesis across wellness domains.
- Built production-ready LLMops/MLOps pipelines using GitHub Actions, Artifact Registry, Secret Manager, Cloud Run, and Cloud Scheduler implementing prompt/config versioning, automated evaluation frameworks, structured observability logging, real-time monitoring, and model performance drift detection for governed AI deployments.
- Implemented end-to-end CI/CD pipelines with automated PR checks (lint, typecheck, tests), manual approval gates, and scalable deployment following Agile/Scrum workflows; set up Firebase Test Lab for automated Android E2E testing and QA automation across the production lifecycle.

Verity Advisors LLC

GenAI Engineer | Remote (USA) | Mar 2025 - Oct 2025

Financial services and fintech firm. Built and deployed ML-driven analytics, LLM-powered explanation systems, and trading infrastructure from prototype to production on AWS.

- Built a real-time, event-driven ML trading system using FastAPI backend services and REST APIs, ingesting live financial data via WebSockets and AWS SQS reducing signal-to-decision latency by approximately 40% in a scalable microservices architecture with sub-500ms inference.
- Developed XGBoost and LSTM models with engineered features (IV Rank, VWAP, RSI, FinBERT/SBERT sentiment) and robust model evaluation/validation frameworks, improving signal precision by approximately 30% and achieving 70% trading accuracy; implemented cost and latency optimization strategies for production serving.
- Architected an AWS-native MLOps pipeline using GitHub Actions, Amazon ECR, EKS/Fargate, S3, MLflow, and Helm/ArgoCD implementing multi-stage CI/CD with data-contract, model-artifact, regression, and staging smoke-test gates, supporting canary-style model promotion and controlled rollouts.

- Implemented production-grade model observability and lifecycle management using CloudWatch, SQS, ElastiCache, EventBridge, and Step Functions enabling inference logging, PSI/KS-based data drift detection, feature-attribution drift monitoring, and automated retraining triggers with structured incident response runbooks.

Wipro Technologies

Data Scientist | India | Jun 2021 - Dec 2024

Global IT services enterprise. Led A/B testing analytics and ML-driven conversion optimization for e-commerce clients.

- Built end-to-end A/B testing and analytics pipelines in BigQuery, defining KPIs (conversion rate, AOV) and performing statistical analysis that contributed to a 15% improvement in conversion rate; collaborated with cross-functional product and engineering teams to translate findings into product decisions.
- Developed a logistic regression conversion model using BigQuery ML with model evaluation and validation, automating batch scoring via Cloud Run; created Power BI dashboards to communicate A/B test results and statistical significance to business stakeholders with clear success metrics.

Wipro Technologies

Data Analyst | India | Oct 2019 - May 2021

Built predictive analytics and dashboards for hospital operations.

- Built ETL pipelines and predictive models (linear regression, random forest) for hospital operations, estimating patient inflow, ICU capacity risk, and staffing demand; created Tableau dashboards enabling real-time KPI monitoring and improving capacity visibility by approximately 10-15%.

PROJECTS

Fraud Monitoring Platform with Graph RAG | [Link](#)

- Developed an XGBoost fraud detection model on 1M+ applications (approximately 1.1% fraud rate, approximately 80% recall) with a Graph RAG-powered agentic explanation layer using LangGraph multi-agent orchestration, LangChain, knowledge graph retrieval, and GPT-4.1 Mini to generate grounded, context-aware fraud explanations.
- Deployed as a production FastAPI microservice on Azure Container Apps with evaluation, guardrails, and cost optimization.

Multimodal Deep Learning for Diet Monitoring | [Link](#)

- Engineered a multimodal Agentic AI system for real time dietary carbohydrate estimation, fusing Bidirectional GRU, frozen ResNet50 (food images), and MLP, achieving RMSE 21.8g and MAE 15.6g on diabetic cohorts; powered by a Google ADK agent autonomously orchestrating MCP-exposed tools across predict, retrieve, explain, and counterfactual reasoning, grounded via Gemini 1.5 Pro + Vertex AI Vector Search RAG to eliminate LLM hallucination in clinical explanations.
- Scaled the system to production on GCP, achieving sub-second mobile inference and zero undetected model drift via BigQuery PSI monitoring and Vertex AI Model Registry versioning; enforced model quality through a multi-threshold eval gate (aggregate, sub-group, operational) before every promotion, and reduced deployment failure risk via GitHub Actions canary CI/CD with automated rollback, delivering an observable, cost-controlled pipeline trusted for diabetic dietary management.

Trip Sage AI Travel Assistant | [Link](#)

- Built an Agentic AI powered conversational travel assistant using Dialogflow CX and Gemini 2.5 Flash (Vertex AI) for generative fallback handling with multiagent webhook workflows, exposing REST API endpoints for flight, hotel, and car rental search with personalized itinerary generation; implemented tool use, function calling, and response evaluation for reliable, grounded outputs.

Multi-Modal Depression Analysis | [Link](#)

- Built a real-time multimodal depression screening system combining audio (openSMILE GeMAPS), computer vision (OpenFace CLNF facial landmarks), and NLP (BERT sentence-transformer) features into a PyTorch deep neural network for clinical prediction; deployed as a scalable FastAPI backend service on Render with Dockerized environments and automated feature extraction pipelines.

Real-Time Electricity Load Forecasting | [Link](#)

- Built deep learning time-series models (LSTM, GRU, BiLSTM) on ERCOT load data with Optuna hyperparameter tuning, reducing forecasting error by approximately 18-22%; deployed as a FastAPI inference service with live Tableau dashboards for real-time grid monitoring.

EDUCATION

M.S. Advanced Data Analytics with AI/ML | University of North Texas, Denton, TX | GPA: 4.0/4.0

B.E. Electronics and Instrumentation | Osmania University, Hyderabad, IN | GPA: 3.7/4.0

CERTIFICATIONS

DeepLearning.AI - Generative AI Specialization